MemoryVault — The Self-Hosted AI Brain for Your Digital Life

The local-first platform that indexes your photos, videos, and documents with AI — so you can search your life in plain English, without ever sending data to the cloud.

Executive Summary

Every human is drowning in digital memories. The average person has 2,000+ photos on their phone, decades of documents scattered across drives, and terabytes of video they'll never watch again — because they can't find anything.

Cloud services like Google Photos and iCloud promise to solve this, but they come with a Faustian bargain: **hand over your most intimate moments to corporations that mine your data for advertising**. The Seattle Shield revelations, the ongoing privacy backlash, and GDPR enforcement have made this trade unacceptable for millions.

MemoryVault is the local-first AI platform that gives you the power of Google Photos' search — "show me beach sunset photos with my daughter from 2023" — running entirely on hardware you own. No cloud. No surveillance. No subscriptions feeding your memories to AI training pipelines.

The Opportunity: Own the $40B+ market for personal data management as privacy becomes the defining consumer technology trend of the decade.

The Problem

Digital Memory Chaos is Universal

- **2.5 trillion photos** taken globally per year — up 25% since 2020
- **Average smartphone** holds 2,400 photos; only 30% are ever viewed again
- **67% of consumers** report "photo/file anxiety" — knowing they have memories but unable to find them
- **5.4 zettabytes** of consumer data created annually — most of it unorganized, unsearchable, forgotten

Cloud Solutions Create New Problems

Issue	Impact
Privacy Exploitation	Google Photos scans every image to build advertising profiles
Data Mining for AI	Apple, Google, Amazon use your photos to train AI models
Surveillance Creep	Seattle Shield revelations show corporate data flows to police
Subscription Lock-in	iCloud charges $130/year for 2TB; Google One $100/year
Vendor Dependency	Your memories held hostage by terms of service changes
No True Deletion	"Deleted" photos persist in cloud backups and training sets

Local Solutions Don't Exist

Current alternatives are painful:

- **Manual organization**: Nobody has 100+ hours to tag a decade of photos

- **Basic NAS search**: Filename-only search is useless for visual content

- **Desktop apps**: Lightroom/Photos.app can't search semantically
- **Self-hosted tools**: PhotoPrism, Immich lack AI-powered natural language search

The gap: No solution offers Google-quality AI search while keeping data 100% local.

Who's Feeling the Pain

- **Privacy-conscious families**: Want to preserve memories without surveillance
- **Photographers/videographers**: Sitting on TBs of footage they can't search (the Maasai Mara problem)
- **Remote workers**: Documents scattered across years of home office chaos
- **Estate planners**: Need to organize digital legacies for families
- **Journalists/activists**: Cannot trust cloud services with sensitive material
- **Healthcare professionals**: Personal data that absolutely cannot leak

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The Solution

MemoryVault: Your Private AI That Knows Everything You've Captured

Core Architecture:

```
                    MemoryVault Core


    Ingest          Vision          Search          Sync
    Engine    →     Index     →     Layer     →     Mesh
                   (Local AI)      (Vector)        (E2EE)

       ↑                                               ↓

             Local Storage (SSD/NAS/External)
    Photos   Videos   Documents   Screenshots   Voice Memos

       ↑

             Device Agents (Mac/PC/Phone/NAS)
```

Core Features

1. Natural Language Search That Actually Works Ask anything: - *"Photos of my dog at the beach"* - *"Videos from Sarah's graduation with grandma in frame"* - *"Documents about the house purchase from 2019"* - *"Screenshots of recipes with pasta"* - *"The sunset photo where I'm wearing the blue jacket"*

How it works: - Vision models (Gemma 4, LLaVA) analyze every image/video frame locally - Whisper transcribes audio in all videos/voice memos - OCR extracts text from documents, screenshots, whiteboards - Vector embeddings enable semantic similarity search - All processing happens on YOUR hardware — nothing leaves

2. Sidecar Metadata Architecture Every file gets a companion `.memoryvault.json`:

```json
{
  "indexed_at": "2026-05-22T08:30:00Z",
  "description": "Golden hour photo at Ocean Beach, San Francisco. Two people visible: adult female (Sa
  "keywords": ["beach", "sunset", "dog", "golden retriever", "San Francisco", "Ocean Beach", "golden hou
  "people": ["Sarah (0.94)"],
  "location": {"lat": 37.7599, "lon": -122.5107, "place": "Ocean Beach, San Francisco, CA"},
  "technical": {"orientation": "landscape", "quality": 0.87, "lighting": "golden hour", "composition":
  "embedding": [0.023, -0.157, ...]
}
```

Why sidecars matter: - Your metadata travels with your files — survives app changes, migrations, disasters - Plain text, grep-able, future-proof - No central database to corrupt or lose - Works with any backup system

3. Face Recognition That Stays Private

- ArcFace embeddings stored locally
- Train it once: "This is Mom, this is Dad, this is my daughter Emma"
- Cross-archive person search: "All photos of Emma since birth"
- Automatic album generation: "Create an album of Emma for her 18th birthday"
- **Zero cloud dependency** — your family's faces never leave your network

4. Multi-Device Sync Without Cloud **The Mesh:** - Phone → Mac → NAS sync with end-to-end encryption - Zero-knowledge relays (optional) for devices behind NAT - Conflict resolution that preserves both versions - Selective sync: keep full archive on NAS, optimized copies on phone - Works offline — sync when devices reconnect

5. Smart Organization Beyond search: - **Auto-albums**: Trips detected by GPS clustering, events by date proximity - **Duplicates**: Perceptual hashing finds near-duplicates across formats - **Curation AI**: Suggests best shots from burst sequences - **Memory surfacing**: "On this day" without sending your data to Apple

6. Multi-Format Intelligence

Format	Processing
Photos	Scene, objects, faces, text (OCR), aesthetic quality
Videos	Frame sampling, motion detection, transcription, speaker ID
Documents	Full-text OCR, layout analysis, document type classification
Voice memos	Transcription, speaker diarization, topic extraction
Screenshots	UI element detection, text extraction, app identification
PDFs	Text extraction, image analysis, table detection

Technology Deep Dive

Local AI Stack

Vision Model Options:

Model	Hardware Requirement	Speed	Quality
Gemma 4 31B (Q4)	32GB RAM, M1+	15 clips/hour	Excellent
LLaVA 13B	16GB RAM	40 clips/hour	Very Good

Model	Hardware Requirement	Speed	Quality
MobileVLM	8GB RAM / Phone	100 clips/hour	Good

Transcription: - WhisperX with speaker diarization - 97 language support - Word-level timestamps for video search

Vector Search: - Local Qdrant or SQLite-VSS - Sub-100ms search across millions of embeddings - Hybrid search: vector similarity + keyword filters

Hardware Compatibility

Supported Platforms: - macOS (Apple Silicon native, Intel supported) - Windows 10/11 (NVIDIA GPU accelerated) - Linux (desktop + headless NAS) - Synology/QNAP NAS (Docker) - Raspberry Pi 5 (limited, photos only) - Android/iOS (mobile agents for sync + capture)

Recommended Specs: - **Minimum**: M1 Mac / 16GB RAM PC / Modern NAS - **Optimal**: M2+ Mac / 32GB RAM + RTX 3080+ PC - **Power User**: Dedicated indexing server, multi-GPU

Privacy Guarantees

What we promise: - **Zero telemetry**: No usage data, no analytics, no phone-home - **Air-gap capable**: Works without any internet connection - **Verifiable**: Open-source core, reproducible builds - **No accounts required**: Use without registration - **Local-only licensing**: License keys work offline forever

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Market Opportunity

Total Addressable Market

Segment	Size	Our Share (5yr)
Photo Management Software	$8B	$1.2B
Personal Cloud Storage	$15B	$2B (migration)
NAS/Home Server Market	$12B	$1.5B (bundling)
Privacy Tech Consumer	$6B	$800M
Total	**$41B**	**$5.5B**

Market Timing

Why Now:

1. **Local AI is finally viable**: Gemma 4 runs on 5-year-old MacBooks (HN front page, May 2026)
2. **Privacy backlash cresting**: Seattle Shield, AI training scandals, GDPR enforcement
3. **Cloud fatigue**: Subscription exhaustion hitting consumers
4. **Hardware ready**: Apple Silicon, modern GPUs make local AI fast
5. **Open models maturing**: Vision-language models matching cloud quality

Competitive Landscape

Competitor	Model	Our Advantage
Google Photos	Cloud, ad-funded	Privacy, ownership, no data mining
Apple Photos	Locked ecosystem	Cross-platform, open, powerful search
Immich	Self-hosted	Better AI, easier setup, sync mesh

Competitor	Model	Our Advantage
PhotoPrism	Self-hosted	Natural language search, video, documents
Synology Photos	NAS-locked	Universal, better AI, mobile-first

Moat building: - Best-in-class local AI pipeline (hard to replicate) - Sidecar standard (network effects as files shared with metadata) - Face embedding library (grows with usage, stays private) - Hardware partnerships (pre-installed on NAS devices)

Business Model

Revenue Streams

1. Software Licensing (Core)

Tier	Price	Features
Personal	$99 one-time	1 user, unlimited devices, core features
Family	$199 one-time	6 users, shared library, family albums
Lifetime	$299 one-time	All future versions, priority support

Why one-time pricing: - Aligns with our privacy promise (no recurring relationship needed) - Differentiates from subscription-fatigued cloud services - Creates immediate value perception - Upsell through hardware partnerships

2. Hardware Partnerships

- **Pre-installed on NAS**: Revenue share with Synology, QNAP, TerraMaster
- **MemoryVault Box**: Dedicated appliance ($599, 4TB included)
- **GPU cloud fallback**: Optional paid processing for underpowered devices

3. Enterprise/Pro

Product	Price	Target
Studio	$499/seat	Photographers, videographers
Team	$999/5 users	Small businesses
Enterprise	Custom	Regulated industries

Unit Economics

- **CAC**: $25 (organic + community)
- **LTV**: $180 (Personal) / $320 (Family)
- **Gross Margin**: 92% (software)
- **Hardware Margin**: 35% (MemoryVault Box)

5-Year Financial Projections

Year	Users	Revenue	Growth
2027	50K	$6M	—
2028	250K	$35M	483%

Year	Users	Revenue	Growth
2029	800K	$110M	214%
2030	2M	$280M	155%
2031	4.5M	$600M	114%

Go-To-Market Strategy

Phase 1: Community Launch (Months 1-6)

Target: Technical Privacy Advocates

- Open-source core on GitHub
- Launch on Hacker News, Reddit r/selfhosted, r/privacy
- YouTube demos: "I indexed 10 years of photos in one night"
- Partner with privacy YouTubers (Techlore, The Hated One)
- Build Discord community (target: 10K members)

Success Metrics: - 5,000 GitHub stars - 10,000 downloads - 500 paying customers

Phase 2: Mainstream Bridge (Months 6-18)

Target: Privacy-Conscious Families

- Polished desktop apps (Mac App Store, Microsoft Store)
- One-click NAS installers
- "Escape Google Photos" migration wizard
- Family sharing features
- Mobile apps for capture + browse

Partnerships: - NAS manufacturers for bundling - Privacy-focused tech reviewers - Photo/video creator communities

Phase 3: Mass Market (Months 18-36)

Target: Everyone Who's Tired of Subscriptions

- MemoryVault Box hardware launch
- Best Buy / Amazon retail presence
- "Your photos, your rules" advertising campaign
- Holiday gift positioning
- International expansion

Team Requirements

Founding Team (First 12 Months)

Role	Focus
CEO/Product	Vision, fundraising, community
CTO	AI/ML architecture, local inference
Lead Engineer	Cross-platform apps, sync protocol
DevRel	Open source community, documentation

Key Hires (12-24 Months)

- Mobile team (iOS + Android)
- Hardware engineer (MemoryVault Box)
- Growth marketer (D2C)
- Enterprise sales lead

Advisors Needed

- Former Apple/Google Photos PM
- Privacy advocacy leader
- NAS industry veteran
- Consumer hardware distribution expert

Funding Strategy

Seed Round: $4M

Use of Funds: - Core product development (60%) - Community building (20%) - Initial marketing (10%) - Operations (10%)

Target Investors: - Privacy-focused VCs (Index, Lux Capital) - Consumer prosumer funds - Angel: privacy advocates with capital

Series A: $20M (Month 18)

Milestone Triggers: - 100K+ active users - $3M+ ARR equivalent - Hardware partnership signed - Mobile apps launched

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Risk Mitigation

Risk	Mitigation
Apple/Google copy features	Open-source moat, cross-platform, privacy commitment they can't make
Local AI too slow	Multiple model tiers, optional cloud fallback, hardware optimization
Setup too complex	One-click installers, migration wizards, NAS pre-installs
Privacy isn't enough value	Leading search quality, not just privacy feature
Hardware costs too high	Software-only focus initially, hardware as premium option

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Why This Team / Why Now

The Perfect Storm

1. **Technical Feasibility**: Local vision-language models just crossed the usability threshold — May 2026 marks the inflection point where Gemma 4 runs on consumer hardware
2. **Market Readiness**: Privacy sentiment at all-time highs post-Seattle Shield, AI training consent debates
3. **Competitive Gap**: No one owns the "local-first AI photo/video search" category
4. **Business Model Clarity**: One-time pricing as rebellion against subscription fatigue

What We're Building

The anti-cloud cloud. All the magic of AI-powered search, none of the surveillance. Your memories stay yours — searchable, organized, private, forever.

Appendix

Technical Specifications

Indexing Performance (M2 Max, 64GB RAM): - Photos: 1,000/hour - Video clips (30s avg): 50/hour - Documents: 500/hour - Full 10-year archive (50K items): ~20 hours (overnight)

Search Performance: - Vector search: <50ms for 1M items - Hybrid search: <100ms - Face search: <200ms

Storage Overhead: - Sidecar metadata: ~2KB per item - Embeddings: ~4KB per item - Thumbnail cache: ~50KB per item - 50K items 3GB overhead

Integration Roadmap

Integration	Timeline	Purpose
Lightroom	Q3 2026	Photographer workflow
Obsidian	Q4 2026	Knowledge management
Home Assistant	Q4 2026	Smart home
Notion	Q1 2027	Document sync
DaVinci Resolve	Q1 2027	Video editing pipeline

"The best camera is the one you have with you. The best photo manager is the one that doesn't spy on you."

MemoryVault: Search your life. Keep your privacy.